THE FITTING IDEALS OF THE SHEAF OF DIFFERENTIALS COMMUTE WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Omega_{X/S} is a finite type quasi-coherent module for f locally of finite type. Morphisms, Lemma morphisms-lemma-finite-type-differentials, used at proof lines 1578-1580. Makes the Fitting ideals defined.
- Base change of differentials. Morphisms, Lemma morphisms-lemma-base-change-differentials, used at proof lines 1580-1583. Identifies Omega_{X'/S'} with the pullback.
- Fitting ideals behave under pullback. used at proof lines 1583-1585. Gives the equality of Fitting ideals after base change.

Source reference. Original lemma 01V0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2429–2429. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01V2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2431–2431. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $X = Z_{-1} \supset Z_0 \supset Z_1 \supset \ldots$ be the closed subschemes defined by the fitting ideals of $\Omega_{X/S}$. Then the formation of Z_i commutes with arbitrary base change.*

Proof. Observe that $\Omega_{X/S}$ is a finite type quasi-coherent $\mathcal{O}_X$-module (Morphisms, Lemma 01V2) hence the fitting ideals are defined. If $f' : X' \to S'$ is the base change of f by $g : S' \to S$, then $\Omega_{X'/S'} = (g')^*\Omega_{X/S}$ where $g' : X' \to X$ is the projection (Morphisms, Lemma 01V0). Hence $(g')^{-1}\mathrm{Fit}_i(\Omega_{X/S}) \cdot \mathcal{O}_{X'} = \mathrm{Fit}_i(\Omega_{X'/S'})$. This means that

$$Z'_i = (g')^{-1}(Z_i) = Z_i \times_X X'$$

scheme theoretically and this is the meaning of the statement of the lemma. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.